

The main idea of the Las Vegas algorithm used here [1] is to perform several trials (10 by default, maximum 30 can be implemented in one cycle) of optimization for different splits into an active training set, a passive training set, a calibration set, and a validation set. In this case, the best split for the calibration set is remembered. Below is an approximate scheme for using the Las Vegas algorithm for compounds with some biological activity. *We thank Professor Veselinovic for the idea to use the Las Vegas algorithm for the CORAL software.*

Run the program (CORALSEA.exe) and click where the pointer is

The screenshot displays the CORAL software interface, titled "CORAL: Please load method or make a random split". The interface is divided into several sections:

- Left Panel:** Contains three vertically stacked plots, each with "MODEL" on the Y-axis and "EXPERIMENT" on the X-axis. The plots are labeled "Active Training set (A)", "Passive Training set (P)", and "Calibration set (C)".
- Top Center:** A "Load method" button and a text field for "Method.txt".
- Center:** A large blue arrow points upwards from the "Las Vegas algorithm" button (which has a value of "10" next to it) to the "Split #TotalSet.txt into training and validation sets" button.
- Right Panel:** Contains various settings and options:
 - SMILES/quasi-SMILES on A,P,C sets:** Includes checkboxes for GRAPH, HSG, HFG, GAO, and a grid for e0, e1, e2, e3, p2, p3, p4, s2, s3, nn.
 - SMILES:** Includes checkboxes for Chaos, s, ss, sss, BOND, NOSP, HALO, and HARD.
 - Atoms pairs proportions:** A table for Cl, Br, N, O, S, P, =, #.
 - Contributions:** Checkboxes for N, O, S, F, Cl, Br, =, #, xyx, xyxx, xyzyx.
 - Classification model:** Radio buttons for CLASSIC SCHEME and CORRELATION BALANCE.
 - Weights:** Input fields for dR weight, IIC weight, CII weight, and CCCP weight.
 - Other parameters:** Input fields for D start, d limit, Nepoch, Threshold start, Threshold maximal, and Number of probes.
- Bottom:** A table for "Split Info" with columns W%, N111, N110, N101, N100, Nall, Defect, and Density. Below this are checkboxes for "Search for duplicates in SMILES" and "Search for duplicates in CAS (ID)".
- Bottom Right:** A "Model Details.txt" button, a date field "So Las Vegas June3, 2024", a text field for "Number of promoters increase/decrease", checkboxes for "STOP and SAVE current CWs", "DemoDCW", and "EvolutionCorr", and an "EXIT" button.

click where the pointer is

CORAL: Please load method or make a random split

☐ Don't show optimization

MODE ↑

Load method Method.txt

Preparation of split April 13, 2022

The first action is "Load". File "#TotalSet.txt" must exist in your folder.

The second action should be "Do distribution".

The third action should be "Save files".

Input Output

1. #TotalSet.txt Load #TotalSet'.txt xxx

2. Do distribution

3. Save files

Dispersion Limit = 0,01

		Planned Distribution	Frequency	Actual Distribution
#TrainingSet.txt	+	0,25	0	0
Invisible training set	-	0,25	0	0
Calibration set	#	0,25	0	0
#ValidationSet.txt	*	0,25	0	0

Loading of vector of Distribution Save vector of Distribution EXIT

W% N111 N110 N101 N100 Nall Defect Density

Split Info 0 0 0 0 0 0 0 0

☐ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

SMILES/quasi-SMILES on A,P,C sets

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

☐ R3 ☐ R4 ☐ R5 ☐ R6 ☐ R7

☐ Chaos ☐ Chaos

☐ SMILES ☐ s ☐ ss ☐ sss

☐ BOND ☐ NOSP ☐ HALO ☐ HARD

Atoms pairs proportions

Cl Br N O S P = #

F... ☐ ☐ ☐ ☐ ☐ ☐

Cl... ☐ ☐ ☐ ☐ ☐ ☐

Br... ☐ ☐ ☐ ☐ ☐ ☐

N... ☐ ☐ ☐ ☐ ☐ ☐

O... ☐ ☐ ☐ ☐ ☐ ☐

S... ☐ ☐ ☐ ☐ ☐ ☐

P... ☐ ☐ ☐ ☐ ☐ ☐

=... ☐ ☐ ☐ ☐ ☐ ☐

All

Contributions

☐ N ☐ O ☐ S

☐ F ☐ Cl ☐ Br

☐ = ☐ #

☐ xyx ☐ xyxx ☐ xyzyx

☐ Classification model

☒ CLASSIC SCHEME

☒ CORRELATION BALANCE

☐ Index of Ideality of Correlation

☐ Correlation Intensity Index

☐ Conformism coefficient

dR weight ☐

IICweight ☐

CIIweight ☐

CCCPweight ☐

D_start ☐ d_limit ☐ Nepoch ☐

Threshold start ☐

Threshold maximal ☐

Number of probes ☐

Model Details.txt So Las Vegas June 3, 2024

Number of promoters increase/decrease ☐

☐ DemoDCW

STOP and SAVE current CW/s ☐ EvolutionCorr

EXIT

click where the pointer is

CORAL: Please load method or make a random split

Don't show optimization

MODE

Load method

Method.txt

Preparation of split April 13, 2022

The first action is "Load". File "#TotalSet.txt" must exist in your folder.

The second action should be "Do distribution".

The third action should be "Save files".

Input Output

+1. #TotalSet.txt Load #TotalSet.txt N = 295

2. Do distribution

3. Save files

Dispersion Limit = 0.005

		Planned Distribution	Frequency	Actual Distribution
#TrainingSet.txt	+	0.250	0	0
Invisible training set	-	0.250	0	0
Calibration set	#	0.250	0	0
#ValidationSet.txt	*	0.250	0	0

Loading of vector of Distribution Save vector of Distribution EXIT

to training and validation sets

s algorithm

10

SMILES/quasi-SMILES on A,P,C sets

CORALSEA-2024v

GRAPH HSG HFG GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

R3 R4 R5 R6 R7

SMILES

Chaos s

Chaos ss

sss

BOND

NOSP

HALO

HARD

Atoms pairs proportions

Cl Br N O S P = #

F..

Cl...

Br.....

N.....

O.....

S.....

P.....

=.....

All

Contributions

N O S

F Cl Br

= #

xyx xyxx xyzyx

Classification model

CLASSIC SCHEME

CORRELATION BALANCE

Index of Ideality of Correlation

Correlation Intensity Index

Conformism coefficient

dR weight

IIC weight

CII weight

CCCP weight

D start d limit Nepoch

Threshold start

Threshold maximal

Number of probes

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease

STOP and SAVE current CWs

DemoDCW

EvolutionCorr

EXIT

W% N111 N110 N101 N100 Nall Defect Density

Split Info

0 0 0 0 0 0 0 0

Search for duplicates in SMILES

Search for duplicates in CAS (ID)

click where the pointer is

CORAL: Please load method or make a random split

Don't show optimization

MODE ↑

Load method

Method.txt

Preparation of split April 13, 2022

The first action is "Load". File "#TotalSet.txt" must exist in your folder.

The second action should be "Do distribution".

The third action should be "Save files".

Input Output

+1. #TotalSet.txt Load #TotalSet.txt N = 295

+2. Do distribution

3. Save files

Dispersion Limit = 0.005

		Planned Distribution	Frequency	Actual Distribution
#TrainingSet.txt	+	0.250	74	0.251
Invisible training set	-	0.250	75	0.254
Calibration set	#	0.250	73	0.247
#ValidationSet.txt	*	0.250	73	0.247

Loading of vector of Distribution Save vector of Distribution EXIT

to training and validation sets

s algorithm 10

SMILES/quasi-SMILES on A,P,C sets

CORALSEA-2024v

GRAPH HSG HFG GAO SMILES

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn R3 R4 R5 R6 R7

Chaos s ss sss BOND NOSP HALO HARD

Atoms pairs proportions Cl Br N O S P = #

Contributions

N O S F Cl Br = # xyx xyxx xyzyx

Classification model CLASSIC SCHEME CORRELATION BALANCE

Index of Ideality of Correlation IICweight

Correlation Intensity Index CIIweight

Conformism coefficient CCCPweight

d_start d_limit Nepoch Threshold start Threshold maximal Number of probes

Model Details.txt So Las Vegas June 3, 2024

Number of promoters increase/decrease

STOP and SAVE current CWs DemoDCW EvolutionCorr EXIT

Search for duplicates in SMILES Search for duplicates in CAS (ID)

click where the pointer is

The screenshot displays the CORAL software interface. A dialog box titled "Preparation of split April 13, 2022" is open, showing the steps for preparing the data split. The main application window is visible in the background, showing the "SMILES/quasi-SMILES on A,P,C sets" section with various checkboxes for selecting features and methods.

Preparation of split April 13, 2022

The first action is "Load". File "#TotalSet.txt" must exist in your folder.

The second action should be "Do distribution".

The third action should be "Save files".

Input: #TotalSet.txt Load Output: #TotalSet'.txt N = 295

+2. Do distribution

+3. Save files

Dispersion Limit = 0.005

		Planned Distribution	Frequency	Actual Distribution
#TrainingSet.txt	+	0.250	74	0.251
Invisible training set	-	0.250	75	0.254
Calibration set	#	0.250	73	0.247
#ValidationSet.txt	*	0.250	73	0.247

Loading of vector of Distribution Save vector of Distribution EXIT

SMILES/quasi-SMILES on A,P,C sets

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nm

☐ R3 ☐ R4 ☐ R5 ☐ R6 ☐ R7

☐ Chaos ☐ Chaos

☐ SMILES ☐ s ☐ ss ☐ sss ☐ BOND ☐ NOSP ☐ HALO ☐ HARD

Atoms pairs proportions
Cl Br N O S P = #

F...
Cl...
Br...
N...
O...
S...
P...
=...

Contributions

☐ N ☐ O ☐ S
☐ F ☐ Cl ☐ Br
☐ = ☐ #
☐ xyx ☐ xyyx ☐ xyzyx

☐ Classification model
☐ CLASSIC SCHEME
☒ CORRELATION BALANCE

☐ Index of Ideality of Correlation
☐ Correlation Intensity Index
☐ Conformism coefficient

dR weight
IIC weight
CII weight
CCCP weight

D_start d_limit Nepoch
Threshold start
Threshold maximal
Number of probes

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease

☐ DemoDCW ☐ EvolutionCort

STOP and SAVE current CW/s EXIT

Search for duplicates in SMILES Search for duplicates in CAS (ID)

Split Info W% N111 N110 N101 N100 Nall Defect Density

0 0 0 0 0 0 0 0

click where the pointer is

 CORAL: select Phase (1 or 2); or change and save method; or import current model

Show optimization **DotSize-1**

MODEL Y↑

→ X

Active Training set (A) EXPERIMENT

...

MODEL Y↑

→ X

Passive Training set (P) EXPERIMENT

...

MODEL Y↑

→ X

Calibration set (C) EXPERIMENT

...

Phase 1: Search for preferable model (T*,N*)

Load method Save method Method.txt

Split #TotalSet.txt into training and validation sets

Phase 2: Building up preferable model (T*,N*)

Las Vegas algorithm 10

Import of current model

W%	N111	N110	N101	N100	Nall	Defect	Density
Split Info	0	0	0	0	0	0	0

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

SMILES/quasi-SMILES on A,P,C sets

#TrainingSet.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

☐ R3 ☐ R4 ☐ R5 ☐ R6 ☐ R7

☒ SMILES

☐ Chaos ☒ s

☐ Chaos ☒ ss

☐ BOND

☐ NOSP

☐ HALO

☐ HARD

Atoms pairs proportions

	Cl	Br	N	O	S	P	=	#
F...	0	0	0	0	0	0	0	0
Cl...	0	0	0	0	0	0	0	0
Br...	0	0	0	0	0	0	0	0
N...	0	0	0	0	0	0	0	0
O...	0	0	0	0	0	0	0	0
S...	0	0	0	0	0	0	0	0
P...	0	0	0	0	0	0	0	0
=...	0	0	0	0	0	0	0	0

Contributions

☐ N ☐ O ☐ S

☐ F ☐ Cl ☐ Br

☐ = ☐ #

☐ xyx ☐ xyxx ☐ xyzyx

☐ Classification model

☐ CLASSIC SCHEME

☒ CORRELATION BALANCE dR weight 0.1

☐ Index of Ideality of Correlation

☐ Correlation Intensity Index

☒ Conformism coefficient CCCP_C 0.25 0

D_{start} 0.5 d_{limit} 0.1 Nepoch 15

Threshold start 3

Threshold maximal 3

Number of probes 1

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease 15

STOP and SAVE current CWs ☒ DemoDCW ☒ EvolutionCorr **EXIT**

click where the pointer is

CORAL: select Phase (1 or 2); or change and save method; or import current model

Show optimization **DotSize-1**

MODEL Y↑

Active Training set (A) EXPERIMENT X→

Passive Training set (P) EXPERIMENT X→

Calibration set (C) EXPERIMENT X→

Load method Save method Method.txt

Split #TotalSet.txt into training and validation sets

Start Las Vegas Calculations 10

APCV style

Import of current model

W%	N111	N110	N101	N100	Nall	Defect	Density
Split Info	0	0	0	0	0	0	0

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

SMILES/quasi-SMILES on A,P,C sets

#TotalSet.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

R3 R4 R5 R6 R7

☒ SMILES

☐ Chaos ☒ s

☐ Chaos ☒ ss

☐ BOND

☐ NOSP

☐ HALO

☐ HARD

Atoms pairs proportions

	Cl	Br	N	O	S	P	=	#
F...	0	0	0	0	0	0	0	0
Cl...	0	0	0	0	0	0	0	0
Br...	0	0	0	0	0	0	0	0
N...	0	0	0	0	0	0	0	0
O...	0	0	0	0	0	0	0	0
S...	0	0	0	0	0	0	0	0
P...	0	0	0	0	0	0	0	0
=	0	0	0	0	0	0	0	0

Contributions

☐ N ☐ O ☐ S

☐ F ☐ Cl ☐ Br

☐ = ☐ #

☐ xyx ☐ xyxx ☐ xyzyx

☐ Classification model

☐ CLASSIC SCHEME

☒ CORRELATION BALANCE dR weight 0.1

☐ Index of Ideality of Correlation

☐ Correlation Intensity Index

☒ Conformism coefficient CCCPC 0.25 0

D_start 0.5 d_limit 0.1 N_epoch 15

Threshold start 3

Threshold maximal 3

Number of probes 10

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease 15

☒ DemoDCW

☒ EvolutionCorr

EXIT

click where the pointer is

Please wait until the calculation is finished

click where the pointer is

CORAL: Wait please...

Show optimization | DotSize=1

Active Training set (A)

Passive Training set (P)

Calibration set (C)

Load method | Save method | Method.txt

Status of the calculations:

Threshold.....3 to 3
Epoch.....16 to 15
Probe.....2 to 10

Las Vegas Calculations in process [10]

APCV style

W%	N111	N110	N101	N100	Nall	Defect	Density
Split Info	98	54	1	0	0	55	45.22 / 0.556

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

SMILES/quasi-SMILES on A,P,C sets

#TotalSet'.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO ☒ SMILES

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn ☐ R3 ☐ Chaos ☒ \$

e0 ☐ R4 ☐ Chaos ☒ ss

e1 ☐ R5 ☐ BOND ☒ sss

e2 ☐ R6 ☐ NOSP

e3 ☐ R7 ☐ HALO

p2 ☐ HARD

p3 **Atoms pairs proportions**

p4 Cl Br N O S P = #

s2 F... 0 0 0 0 0 0 0

s3 Cl.... 0 0 0 0 0 0 0

nn Br..... 0 0 0 0 0 0 0

Contributions

☐ N ☐ O ☐ S N..... 0 0 0 0 0

☐ F ☐ Cl ☐ Br O..... 0 0 0 0 0

☐ = ☐ # S..... 0 0 0 0 0

☐ xyx ☐ xyyx ☐ xyzyx P..... 0 0 0

☐ Classification model =..... 0

☐ CLASSIC SCHEME All

☒ CORRELATION BALANCE dR weight 0.1

☐ Index of Ideality of Correlation

☐ Correlation Intensity Index

☒ Conformism coefficient CCCPC 0.25 0

D_start 0.5 d_limit 0.1 N_epoch 15

Threshold start 3

Threshold maximal 3

Number of probes 10

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease 15

☒ DemoDCW ☒ EvolutionCorr EXIT

click where the pointer is
click where the pointer is

click where the pointer is

The screenshot displays the CORAL software interface. At the top, a status bar reads "CORAL: Wait please...". The main window is titled "Show optimization" and features a "DotSize-1" button. It contains three vertically stacked plots, each with "MODEL" on the Y-axis and "EXPERIMENT" on the X-axis. The plots are labeled "Active Training set (A)", "Passive Training set (P)", and "Calibration set (C)". To the right of the plots, there are buttons for "Load method", "Save method", and a text field for "Method.txt". Below these, a status message reads: "Status of the calculations: Threshold.....3 to 3 THE CALCULATION IS COMPLETED Probe.....10 to 10". A prominent dialog box titled "Las Vegas Calculations in process" is centered, showing a progress bar at 10 and the text "APCV style". A blue arrow points to the "OK" button in this dialog. In the background, another window titled "CORALSEA-2024lv" is visible, showing various settings for SMILES, Chaos, and Atoms pairs proportions. At the bottom, there is a table for "Split Info" with columns for W%, N111, N110, N101, N100, Nall, Defect, and Density. The "Defect" value is 49.14 and "Density" is 0.0000. Other settings at the bottom include "Search for duplicates in SMILES", "Search for duplicates in CAS (ID)", "DemoDCW", "EvolutionCorr", and an "EXIT" button.

click where the pointer is

run the program and click where the pointer is

click where the pointer is

click where the pointer is

add 'LV' to the file name and click where the pointer is

click where the pointer is

CORAL: Please load method or make a random split

Don't show optimization

M O D E ↑

Preparation of split April 13, 2022

The first action is "Load". File "#TotalSet.txt" must exist in your folder.

The second action should be "Do distribution".

The third action should be "Save files".

	Input	Output
1.	#TotalSet.txt → Load	#TotalSet'.txt
2.	Do distribution	
3.	Save files	

Dispersion Limit = 0.01

		Planned Distribution	Frequency	Actual Distribution
#TrainingSet.txt	+	0.25	0	0
Invisible training set	-	0.25	0	0
Calibration set	#	0.25	0	0
#ValidationSet.txt	*	0.25	0	0

Loading of vector of Distribution Save vector of Distribution EXIT

W% N111 N110 N101 N100 Nall Defect Density
Split Info 0 0 0 0 0 0 0

Search for duplicates in SMILES Search for duplicates in CAS (ID)

Method.txt

to training and validation sets

s algorithm 10

SMILES/quasi-SMILES on A,P,C sets

CORALSEA-2024v

GRAPH HSG HFG GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

R3 R4 R5 R6 R7

Chaos Chaos

Atoms pairs proportions
Cl Br N O S P = #

F... Cl... Br... N... O... S... P... =...

Contributions

N O S
F Cl Br
= #
xyx xyyx xyzyx

Classification model

CLASSIC SCHEME

CORRELATION BALANCE

Index of Ideality of Correlation IIC weight
Correlation Intensity Index CII weight
Conformism coefficient CCCP weight

d start d limit N epoch
Threshold start
Threshold maximal
Number of probes

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease

DemoDCW EvolutionCorr

STOP and SAVE current CW's

EXIT

click where the pointer is

The screenshot displays the CORAL software interface. A dialog box titled "Preparation of split April 13, 2022" is open, showing instructions for file preparation and a table of planned vs. actual distribution frequencies. The main window shows the "CORALSEA-2024lv" interface with various settings for SMILES/quasi-SMILES on A,P,C sets, including checkboxes for different models and a table for atom pairs proportions.

Preparation of split April 13, 2022

The first action is "Load". File "#TotalSet.txt" must exist in your folder.

The second action should be "Do distribution".

The third action should be "Save files".

Input	Output	
+1. #TotalSetLV.t	Load	#TotalSet'.txt N = 295
2.	Do distribution	
+3.	Save files	

Dispersion Limit = 0.005

		Planned Distribution	Frequency	Actual Distribution
#TrainingSet.txt	+	0.250	74	0.251
Invisible training set	-	0.250	75	0.254
Calibration set	#	0.250	73	0.247
#ValidationSet.txt	*	0.250	73	0.247

Buttons: Loading of vector of Distribution, Save vector of Distribution, EXIT

CORALSEA-2024lv

SMILES/quasi-SMILES on A,P,C sets

GAO

SMILES

Chaos

R3

R4

R5

R6

R7

BOND

NOSP

HALO

HARD

Atoms pairs proportions

Cl Br N O S P = #

Contributions

N O S

F Cl Br

= #

xyx xyxx xyzyx

Classification model

CLASSIC SCHEME

CORRELATION BALANCE

Index of Ideality of Correlation

Correlation Intensity Index

Conformism coefficient

CCCPCweight

dR weight

IICweight

CIIweight

CCCPCweight

D_start d_limit Nepoch

Threshold start

Threshold maximal

Number of probes

Model Details.txt

So Las Vegas June 3, 2024

Number of promoters increase/decrease

STOP and SAVE current CWs

DemoDCW

EvolutionCorr

EXIT

W% N111 N110 N101 N100 Nall Defect Density

Split Info

Search for duplicates in SMILES

Search for duplicates in CAS (ID)

 CORAL: select Phase (1 or 2); or change and save method; or import current model

SMILES/quasi-SMILES on A,P,C sets

#TrainingSet.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

	e0	e1	e2	e3	p2	p3	p4	s2	s3	nn
e0	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
e1	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
e2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
e3	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
p2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
p3	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
p4	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
s2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
s3	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
nn	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Contributions

☐ N ☐ O ☐ S
☐ F ☐ Cl ☐ Br
☐ = ☐ #
☒ xyx ☐ xyyx ☐ xyzyx

☐ Classification model
☐ CLASSIC SCHEME
☒ CORRELATION BALANCE

☐ Index of Ideality of Correlation
☐ Correlation Intensity Index

☒ Conformism coefficient CCCPc

D_start 0.5 d_limit 0.1 N_epoch 15
Threshold start 3
Threshold maximal 3
Number of probes 10

dR weight 0.1

Atoms pairs proportions

	Cl	Br	N	O	S	P	=	#
F.....	0	0	0	0	0	0	0	0
Cl.....	0	0	0	0	0	0	0	0
Br.....	0	0	0	0	0	0	0	0
N.....	0	0	0	0	0	0	0	0
O.....	0	0	0	0	0	0	0	0
S.....	0	0	0	0	0	0	0	0
P.....	0	0	0	0	0	0	0	0
=.....	0	0	0	0	0	0	0	0

All

Model Details.txt So Las Vegas June 3, 2024

Number of promoters increase/decrease 15

STOP and SAVE current CWs ☒ DemoDCW ☒ EvolutionCorr EXIT

click where the pointer is

CORAL: Building up preferable model

Show optimization **DotSize-1**

MODEL Y↑

→ X

Active Training set (A) EXPERIMENT

...

Phase 2: Building up preferable model (T*,N*)

Define threshold and press Continue

3 Continue

MODEL Y↑

→ X

Passive Training set (P) EXPERIMENT

...

MODEL Y↑

→ X

Calibration set (C) EXPERIMENT

...

Import of current model

W%	N111	N110	N101	N100	Nall	Defect	Density
Split Info	0	0	0	0	0	0	0

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

SMILES/quasi-SMILES on A,P,C sets

#TrainingSet.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

☐ R3 ☐ R4 ☐ R5 ☐ R6 ☐ R7

☒ SMILES ☐ Chaos ☐ Chaos ☐ s ☐ ss ☐ sss ☐ BOND ☐ NOSP ☐ HALO ☐ HARD

Atoms pairs proportions

	Cl	Br	N	O	S	P	=	#
F	0	0	0	0	0	0	0	0
Cl	0	0	0	0	0	0	0	0
Br	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0
O	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0
=	0	0	0	0	0	0	0	0

Contributions

☐ N ☐ O ☐ S ☐ F ☐ Cl ☐ Br ☐ = ☐ # ☐ xyx ☐ xyxx ☐ xyzyx

☐ Classification model ☐ CLASSIC SCHEME ☒ CORRELATION BALANCE

☐ Index of Ideality of Correlation ☐ Correlation Intensity Index

☒ Conformism coefficient CCCP_C 0,25 0

D_{start} 0,5 d_{limit} 0,1 N_{epoch} 15

Threshold start 3

Threshold maximal 3

Number of probes 10

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease 15

☐ STOP and SAVE current CW/s ☒ DemoDCW/ ☒ EvolutionCorr **EXIT**

Please wait until the calculation is finished and click where the pointer is

 CORAL: you should save model now

Show optimization DotSize-1

MODEL Y↑

Active Training set (A) → X EXPERIMENT

...

MODEL Y↑

Passive Training set (P) → X EXPERIMENT

...

MODEL Y↑

Calibration set (C) → X EXPERIMENT

...

Selected threshold is 3
THE CALCULATION IS COMPLETED

Phase 2: Building up preferable model (T*,N*)

Define threshold and press Continue 3 Continue

C0 = -2.8843809 C1 = 0.3947727

Insert a SMILES for calculation of DCW and EndPoint

Demo of calculation of DCW and endpoint will be in file

Start of DCW and Endpoint calculation for SMILES from file

Save Model

W%	N111	N110	N101	N100	Nall	Defect	Density
Split Info	100	51	0	0	0	51	43.21
							0.543

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

SMILES/quasi-SMILES on A,P,C sets

#TrainingSet.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO ☒ SMILES

☐ e0 ☐ e1 ☐ e2 ☐ e3 ☐ p2 ☐ p3 ☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R3 ☐ Chaos ☒ s

☐ e0 ☐ e1 ☐ e2 ☐ e3 ☐ p2 ☐ p3 ☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R4 ☐ Chaos ☒ ss

☐ e1 ☐ e2 ☐ e3 ☐ p2 ☐ p3 ☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R5 ☐ BOND

☐ e2 ☐ e3 ☐ p2 ☐ p3 ☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R6 ☐ NOSP

☐ e3 ☐ p2 ☐ p3 ☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R7 ☐ HALO

☐ p2 ☐ p3 ☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R7 ☐ HARD

☐ p3 ☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R7 ☐ HARD

☐ p4 ☐ s2 ☐ s3 ☐ nn ☐ R7 ☐ HARD

☐ s2 ☐ s3 ☐ nn ☐ R7 ☐ HARD

☐ s3 ☐ nn ☐ R7 ☐ HARD

☐ nn ☐ R7 ☐ HARD

Contributions

☐ N ☐ O ☐ S

☐ F ☐ Cl ☐ Br

☐ = ☐ #

☐ xyx ☐ xyxx ☐ xyzyx

☐ Classification model

☐ CLASSIC SCHEME

☒ CORRELATION BALANCE dR weight 0.1

☐ Index of Ideality of Correlation

☐ Correlation Intensity Index

☒ Conformism coefficient CCCP_C 0.25 0

D start 0.5 d limit 0.1 Nepoch 15

DemoDCW.txt Threshold start 3

Threshold maximal 3

Number of probes 1

#ValidationSet.txt #ModelForValidationSet.txt

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease 15

☒ DemoDCW

☒ EvolutionCorr

STOP and SAVE current CW's EXIT

click where the pointer is twice 1.Load method; and 2 Import of current model

 CORAL: select Phase (1 or 2); or change and save method; or import current model

Show optimization **DotSize-1**

MODEL Y↑

Active Training set (A) → X EXPERIMENT

...

MODEL Y↑

Passive Training set (P) → X EXPERIMENT

...

MODEL Y↑

Calibration set (C) → X EXPERIMENT

...

Load method **Save method** **Method.txt**

Phase 1: Search for preferable model (T*,N*)

Split #TotalSet.txt into training and validation sets

Phase 2: Building up preferable model (T*,N*)

Las Vegas algorithm **10**

Import of current model ←

W%	N111	N110	N101	N100	Nall	Defect	Density
Split Info	0	0	0	0	0	0	0

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

SMILES/quasi-SMILES on A,P,C sets

#TrainingSet.txt

☐ **GRAPH** ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

☐ R3 ☐ R4 ☐ R5 ☐ R6 ☐ R7

☐ Chaos ☒ **SMILES** ☐ Chaos ☒ \$ ☒ ss ☐ sss

☐ BOND ☐ NOSP ☐ HALO ☐ HARD

Atoms pairs proportions

	Cl	Br	N	O	S	P	=	#
F...	0	0	0	0	0	0	0	0
Cl....	0	0	0	0	0	0	0	0
Br.....	0	0	0	0	0	0	0	0
N.....	0	0	0	0	0	0	0	0
O.....	0	0	0	0	0	0	0	0
S.....	0	0	0	0	0	0	0	0
P.....	0	0	0	0	0	0	0	0
=.....	0	0	0	0	0	0	0	0

Contributions

☐ N ☐ O ☐ S ☐ F ☐ Cl ☐ Br ☐ = ☐ # ☐ xyx ☐ xyxx ☐ xyzyx

☐ **Classification model** ☐ **CLASSIC SCHEME** ☒ **CORRELATION BALANCE** **dR weight** **0.1**

☐ Index of Ideality of Correlation ☐ Correlation Intensity Index

☒ **Conformism coefficient** **CCCP_C** **0.25** **0**

D_{start} **0.5** **d_{limit}** **0.1** **N_{epoch}** **15**

Threshold start **3** **Threshold maximal** **3** **Number of probes** **1**

Model Details.txt **So Las Vegas June3,2024**

Number of promoters increase/decrease **15**

STOP and SAVE current CW/s ☒ **DemoDCW** ☒ **EvolutionCorr** **EXIT**

click where the pointer is to check model on validation set

CORAL: Calculation of model for external substances

Hide DotSize=1

Active Training set (A) EXPERIMENT
n=74; R2=0.5860; s=1.37; MAE=1.19; F=102

Passive Training set (P) EXPERIMENT
n=75; R2=0.6976; s=1.15; MAE=0.983; F=168

Calibration set (C) EXPERIMENT
n=73; R2=0.8655; s=0.708; MAE=0.551; F=457

XY-dots analysis **Marker-0**

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

Phase 1: Search for preferable model [T*,N*]

Split #TotalSet.txt into training and validation sets.

Phase 2: Building up preferable model [T*,N*]

Calculation model for a list of SMILES

list.txt ListModel.txt

C0 = -2.8843809 C1 = 0.3947727

Insert a SMILES for calculation of DCW and EndPoint

Demo of calculation of DCW and endpoint will be in file

DCW(3,15)= EndPoint =

Start of DCW and Endpoint calculation for SMILES from file

Import of current model

W% N111 N110 N101 N100 Nall Defect Density

Split Info 100 51 0 0 0 51 43.21 0.543

Continue optimization

SMILES/quasi-SMILES on A,P,C sets

#TrainingSet.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

☐ R3 ☐ R4 ☐ R5 ☐ R6 ☐ R7

☒ SMILES

☐ Chaos ☒ s

☐ Chaos ☒ ss

☐ BOND

☐ NOSP

☐ HALO

☐ HARD

Atoms pairs proportions

	Cl	Br	N	O	S	P	=	#
F	0	0	0	0	0	0	0	0
Cl	0	0	0	0	0	0	0	0
Br	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0
O	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0
=	0	0	0	0	0	0	0	0
#	0	0	0	0	0	0	0	0

Contributions

☐ N ☐ O ☐ S

☐ F ☐ Cl ☐ Br

☐ = ☐ #

☐ xyx ☐ xyxx ☐ xyzyx

☐ Classification model

☐ CLASSIC SCHEME

☒ CORRELATION BALANCE dR weight 0.1

☐ Index of Ideality of Correlation

☐ Correlation Intensity Index

☒ Conformism coefficient CCCPC 0.25 0

D_start 0.5 d_limit 0.1 Nepoch 15

DemoDCW.txt Threshold start 3

Threshold maximal 3

Number of probes 1

#ValidationSet.txt #ModelForValidationSet.txt

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease 15

☒ DemoDCW

☒ EvolutionCorr

EXIT

STOP and SAVE current CW's

Please see results for validation set in file .../model/#ModelForValidationSet.txt; Now you can study plots "expr vs calc"

Hide

DotSize-1

Active Training set (A)

MODEL vs EXPERIMENT

n=74: R2=0.5860; s=1.37; MAE=1.19; F=102

Passive Training set (P)

MODEL vs EXPERIMENT

n=75: R2=0.6976; s=1.15; MAE=0.983; F=168

Calibration set (C)

MODEL vs EXPERIMENT

n=73: R2=0.8655; s=0.708; MAE=0.551; F=457

Validation set (V)

MODEL vs EXPERIMENT

n=73: R2=0.8377; s=0.776; MAE=0.589; F=367

XY-dots analysis **Marker-0**

☒ Search for duplicates in SMILES ☐ Search for duplicates in CAS (ID)

Phase 1: Search for preferable model (T*,N*)

Split #TotalSet.txt into training and validation sets

Phase 2: Building up preferable model (T*,N*)

Calculation model for a list of SMILES

list.txt ListModel.txt

C0 = -2.8843809 C1 = 0.3947727

Insert a SMILES for calculation of DCW and EndPoint

c1(ccc([N])(=O)O)c(c1)Cl

Demo of calculation of DCW and endpoint will be in file

DCW(3,15)= 14.2090 EndPoint =

Start of DCW and Endpoint calculation for SMILES from file

Import of current model

W%	N111	N110	N101	N100	Nall	Defect	Density
100	51	0	0	0	51	43.21	0.543

Split Info

☒ Continue optimization

SMILES/quasi-SMILES on A,P,C sets

#TrainingSet.txt

☐ GRAPH ☐ HSG ☐ HFG ☐ GAO

e0 e1 e2 e3 p2 p3 p4 s2 s3 nn

☐ R3 ☐ R4 ☐ R5 ☐ R6 ☐ R7

☒ SMILES ☐ Chaos ☐ Chaos ☐ s ☐ ss ☐ sss ☐ BOND ☐ NOSP ☐ HALO ☐ HARD

Atoms pairs proportions

	Cl	Br	N	O	S	P	=	#
F...	0	0	0	0	0	0	0	0
Cl...	0	0	0	0	0	0	0	0
Br...	0	0	0	0	0	0	0	0
N...	0	0	0	0	0	0	0	0
O...	0	0	0	0	0	0	0	0
S...	0	0	0	0	0	0	0	0
P...	0	0	0	0	0	0	0	0
=	0	0	0	0	0	0	0	0

Contributions

☐ N ☐ O ☐ S ☐ F ☐ Cl ☐ Br ☐ = ☐ # ☐ xyx ☐ xyxx ☐ xyzyx

☐ Classification model ☐ CLASSIC SCHEME ☒ CORRELATION BALANCE

dR weight 0.1

☐ Index of Ideality of Correlation ☐ Correlation Intensity Index

☒ Conformism coefficient

CCCPC 0.25 0

D_start 0.5 d_limit 0.1 Nepoch 15

DemoDCW.txt Threshold start 3 Threshold maximal 3 Number of probes 1

2.7249

#ValidationSet.txt #ModelForValidationSet.txt

Model Details.txt So Las Vegas June3,2024

Number of promoters increase/decrease 15

☒ DemoDCW ☒ EvolutionCorr

STOP and SAVE current CW's

EXIT

Reference

1. Toropova A.P., Toropov A.A. The coefficient of conformism of a correlative prediction (CCCP): Building up reliable nano-QSPRs/QSARs for endpoints of nanoparticles in different experimental conditions encoded via quasi-SMILES (2024) Science of the Total Environment, 927, art. no. 172119, DOI: 10.1016/j.scitotenv.2024.172119